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Candidate Rotation Periods for 34 Main-belt Asteroids from Gaia DR3 Sparse Photometry

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ABSTRACT

I searched Gaia DR3 sparse photometry of 598 well-observed numbered asteroids for rotation periods, fitting at each trial frequency a linear model with season offsets, a linear phase-angle term and a two-harmonic Fourier series. On 336 confident results that can be compared with published periods, 80% agree within 3%; accuracy falls to 36% above 50 h, where the method usually returns half of the true period. I list 34 asteroids whose confident period was absent from all major period compilations, as targets for confirmation. An independent reanalysis of Gaia DR3 with a refined method recovers the same period for all 34.

DATA

I ranked the numbered asteroids in the Gaia DR3 `sso_source` table (Tanga et al. 2023) by their number of observations and kept the 6,000 best observed. Going down this list, I took the first 300 that have a rotation period of 2–100 h in JPL SBDB, which follows the October 2023 LCDB (Warner et al. 2009) (*validation set*), and the first 298 that have no JPL period (*target set*).

Gaia reports one G magnitude per CCD crossing. For each transit (`transit_id`) I averaged the epochs and the magnitudes of its CCD observations. The uncertainty σ_i of transit i is the largest relative flux error among its CCDs, converted to magnitudes and added in quadrature to a 0.03 mag floor. JPL Horizons, with Gaia as the observer, gave for each transit the heliocentric distance r_i, the distance Δ_i from Gaia and the phase angle α_i. Epochs were corrected for light time, $t_i \to t_i - \Delta_i/c$. A new *season* starts wherever consecutive transits are more than 60 d apart.

METHOD

For a trial frequency f the reduced magnitude is modelled as

$$G_i - 5\log_{10}(r_i\Delta_i) = c_{s(i)} + \beta\,\alpha_i + \sum_{k=1}^{2}\left[a_k\cos(2\pi k f t_i) + b_k\sin(2\pi k f t_i)\right], \tag{1}$$

where $s(i)$ is the season of transit i. All coefficients are free and enter linearly: one magnitude offset c_s per season, which absorbs the change of viewing geometry and aspect between seasons, a common phase-angle slope β, and four Fourier amplitudes a_1, b_1, a_2, b_2. They are solved by weighted least squares with weights σ_i^{-2}, giving $\chi^2(f)$. Before the search, transits deviating by more than five robust standard deviations (1.4826× median absolute deviation) from the fit without Fourier terms are removed, in two passes.

I evaluated $\chi^2(f)$ on a uniform grid from 0.24 to 12 rotations per day ($P = 1/f$ from 100 to 2 h) with a step of $0.1/T$, where T is the time span of the data. The best frequency f_1 gives the peak period $P_1 = 1/f_1$. Because most lightcurves are double-peaked and the second harmonic can fit them at half the rotation period, I adopt $2P_1$ whenever $\chi^2(f_1/2) < 1.5\,\chi^2(f_1)$ and $2P_1 \le 100$ h. The factor 1.5 was chosen by two-fold cross-validation on the validation set. The amplitude is the peak-to-peak range of the fitted Fourier series.

The confidence score is $S = \ln(\chi_0^2/\chi_1^2)\,(\chi_2^2/\chi_1^2 - 1)$, where χ_0^2 is the fit without Fourier terms, $\chi_1^2 = \chi^2(f_1)$, and χ_2^2 is the lowest χ^2 farther than $5/T$ from f_1, $2f_1$ and $f_1/2$. The first factor measures how much rotation improves the fit, the second how much the best peak stands out. The threshold $S \ge 0.1604$ is the lowest score at which the highest-scoring validation results reach 80% agreement within 3%. The code is archived with the data (Tropynin 2026a).

VALIDATION

The cross-validated recovery rate on the full validation set is 67.7%. Above the threshold:

Table 1. The 34 candidate rotation periods.

Number	P (h)	Amp.	N	Number	P (h)	Amp.	N
5897	6.0729	0.39	67	44204	4.7663	0.72	65
8751	3.2402	0.25	70	45726	7.9396	0.33	69
11621	33.7373	0.60	59	47917	4.7480	0.38	80
11818	2.7435	0.17	66	50588	56.9297[a]	0.55	65
13363	8.9320	0.26	68	52653	3.7362	0.14	65
16916	3.8301	0.15	69	55335	5.0076	0.14	61
17337	4.9700	0.21	65	56994	14.7458	0.39	64
22427	5.4857	0.46	66	63007	4.4476	0.33	65
24359	16.1525	0.22	62	64303	8.8781	0.19	63
24711	6.4557	0.75	61	65213	3.8679	0.34	73
30068	8.3416	0.27	72	72885	3.0007	0.32	62
31611	5.4767	0.38	62	84508	3.0394	0.23	69
32778	66.9411[a]	0.56	63	87249	4.4550	0.19	70
35653	30.0890	0.13	66	88237	52.2814[a]	0.10	58
40172	5.1295	0.54	74	91539	5.6743	0.42	91
42087	81.9684[a]	0.22	66	97578	2.8696	0.41	86
44017	3.0915	0.20	65	105992	8.0425	0.19	58

NOTE—Amp.: peak-to-peak amplitude of the fitted Fourier series (mag). N: number of transits used. [a]Period above 50 h; the true period may be twice the value given.

- validation set: 172 of 214 agree (in-sample, because this set fixed the threshold);
- target-set asteroids that Ďurech & Hanuš (2023) solved from the same Gaia data with a different method: 81 of 97 agree;
- target-set asteroids published since by Cellino et al. (2024), MacLennan et al. (2026) or McNeill et al. (2023): 15 of 25 agree.

In total 268 of 336 (80%) agree. Accuracy is 81–89% below 50 h and 10 of 28 above it, where half the true period is usually returned. Slow rotators beyond 100 h receive spurious short periods.

THE 34 CANDIDATES

Of the 298 target-set asteroids, 165 pass the threshold. I removed the 97 solved by Ďurech & Hanuš (2023) and every asteroid with a rotation period, in October 2026, in JPL SBDB, LCDB V4.0, DAMIT (Ďurech et al. 2010), the SsODNet ssoBFT table (Berthier et al. 2023), about 180 VizieR tables (5.1 million rows), or the Minor Planet Bulletin volumes 50–53. Of the 42 that remained, a reanalysis of all of Gaia DR3 with a refined, independently calibrated method (Tropynin 2026b) dropped seven (four probably at twice the period found here, three now preferring slow solutions), and Ďurech et al. (2022) already gives 19.72 h for 11857 from ATLAS photometry. The remaining 34 (Table 1) are all main-belt asteroids with 58–91 transits in 4–7 seasons, and the reanalysis recovers the same period within 3% for each of them. From the validation, I expect about 83% of the 30 periods below 50 h to be correct. The repository lists for every candidate the alias period, the number of seasons, the time span and the score. I offer the list as targets for dense-lightcurve confirmation.

This work used data from the ESA mission Gaia, processed by the Gaia DPAC; JPL Horizons and the JPL Small-Body Database; and VizieR (CDS, Strasbourg). The analysis was designed, coded and checked with the help of an AI assistant (Claude, Anthropic) under the author's direction.

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